AIORI-3 HACKATHON

Privacy-Aware IPv6 IoT Security Monitoring Across Address Rotation

PS-007 | Cybersecurity • IPv6 • IoT • Privacy

Team	Last Minute Coders
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Submission presentation • October 2026

1. Problem Description

IPv6 IoT devices may legitimately change addresses because of privacy extensions, SLAAC, DHCPv6, or network churn. Monitoring systems that rely on stable IP addresses can lose continuity across rotations. At the same time, collecting raw device identifiers and full addresses can expose sensitive operational information.

- **Security gap:** rapid address changes and reconnect patterns can make suspicious behavior harder to correlate.
- **Privacy gap:** logs containing persistent device identifiers and full IPv6 addresses may reveal more than operators need.
- **Goal:** detect suspicious patterns while returning pseudonymized identities and redacted, explainable alerts.

Scope note: address rotation is not inherently malicious; the prototype flags patterns for review rather than declaring an attack with certainty.

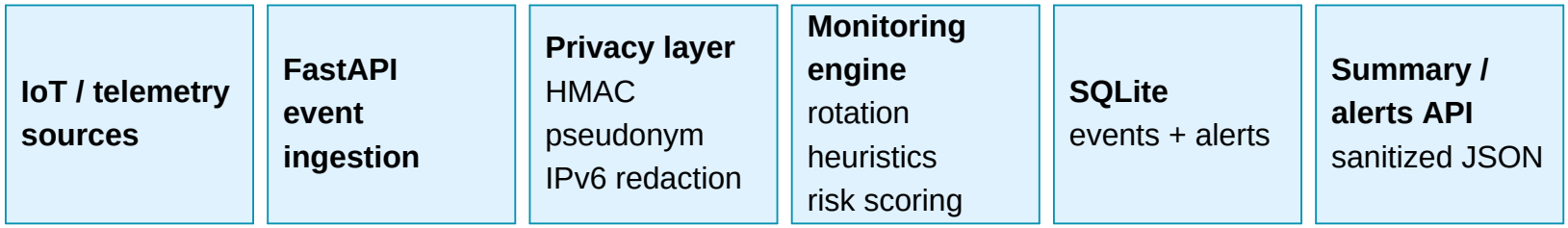
2. Proposed Solution

A lightweight monitoring API ingests IoT telemetry, pseudonymizes device identities, evaluates rotation/reconnect heuristics, stores events and alerts, and exposes privacy-aware summaries.

- **Ingest:** FastAPI endpoint receives device ID, IPv6 address, event type, network segment and related telemetry.
- **Protect:** HMAC-SHA256 creates a stable pseudonymous device key; external-facing summaries redact IPv6 details.
- **Detect:** heuristic checks look for rapid address rotation, repeated reconnects and unusual behavior.
- **Explain:** alerts include a severity/risk score and human-readable reason.
- **Persist:** SQLite stores telemetry history and alert records for the prototype.

Expected output: actionable signals for a human operator without exposing raw device identity in ordinary summaries.

3. Architecture and Data Flow



Key interfaces documented in the project: POST /api/v1/events, GET /api/v1/summary, and GET /api/v1/alerts.

The live deployment and all runtime behavior should be tested before presenting this as a verified production system.

4. Optimization and Privacy by Design

- **Data minimization:** operator-facing responses use pseudonymous device keys and redacted IPv6 details.
- **Stable pseudonyms:** HMAC-based keys help correlate events from the same device without displaying its raw ID.
- **Explainable detection:** rule-based signals can be inspected and tuned by operators.
- **Lightweight deployment:** Python, FastAPI and SQLite reduce setup friction; Docker files support repeatable demos.
- **Safe interpretation:** benign privacy-address rotation can occur; alerts should be triaged with context.

Known limitations to disclose: heuristic thresholds require validation against representative traffic; a demo dataset is not a substitute for a real-world evaluation; protect the HMAC secret and restrict access to raw telemetry.

5. Delivery Timeline and Demo Plan

Stage	Activities	Evidence
Prototype	Implement ingestion, pseudonymization, storage and alert endpoints.	Source code and API examples
Validation	Test normal events, rapid address rotation, redaction and error handling.	Test results / screenshots
Submission prep	Prepare PDF, README, team introduction video and repository checklist.	GitHub deliverables
Demo	Start service, send normal event, send rotation event, inspect summary and alert.	Live demo or recorded walkthrough

Five-minute demo: (1) start API, (2) submit normal telemetry, (3) submit a changed IPv6 address for the same device, (4) fetch alerts/summary, (5) explain pseudonymization and redaction.

6. References and Submission Links

- AIORI-3 Hackathon Guidelines: https://portal.aiori.in/aiori-3/hackathon-guideline/
- Project repository: https://github.com/goyalakrit-cyber/ps-007-privacy-aware-ipv6-iot-security-monitoring
- Official AIORI template: https://github.com/aiori-hackathon/A3-PSXXX-TCXXX
- Python: https://www.python.org/
- FastAPI: https://fastapi.tiangolo.com/
- SQLAlchemy: https://www.sqlalchemy.org/

Before final submission

- Confirm the team repository is created from the official template and named with the team code.
- Add the AIORI GitHub account as collaborator and complete the required ownership transfer.
- Upload this PDF and all source/supporting files to the official team repository.
- Upload a team introduction video to YouTube and add its public/unlisted link to README.
- Add a live demo URL, or include the final deliverables for evaluator review.

Prepared from the project details currently documented in the team's public repository. Verify implementation claims and team details before submission.